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Lab 10 Unsupervised Learning

ITAI-1371

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Reflection on Unsupervised Learning

In this lab, our group explored unsupervised learning and how it differs from the supervised learning methods we studied in previous labs. We worked with K-Means Clustering and Principal Component Analysis (PCA) to understand how machine learning can recognize patterns without target labels. We used a synthetic customer dataset to identify groups of similar data points and the Iris dataset to reduce the number of features. Unlike our previous assignments, where we trained models to predict specific outcomes, this lab encouraged us to focus on finding patterns within the data. We learned that machine learning can provide useful insights even when we do not have correct answers to compare against.

In Task 1, we explored the Elbow Method to determine the optimal number of clusters for K-Means (Tien). We tested values of k from 1 to 10 and created a graph showing how inertia changed as the number of clusters increased (Tien). The graph showed that inertia decreased with more clusters, but the improvement became much smaller after k = 4 (Tien). At first, choosing more clusters might seem better because it produces lower inertia (Tien). However, we learned that too many clusters can split the data into groups that we do not really need (Tien). One challenge was figuring out where the elbow appeared instead of just choosing the lowest inertia value (Tien). By comparing how quickly inertia decreased before and after k = 4, we were able to understand why this value made the most sense for our dataset (Tien). This made us wonder how we would select k if the graph did not show such a clear turning point (Tien).

In Task 2, we applied K-Means with four clusters and created a scatter plot to visualize the results (Viktoriya). We colored the data points according to their assigned clusters and marked the centroids to show the center of each group (Viktoriya). Seeing the clusters on the graph helped us understand how K-Means groups nearby points based on their similarities (Viktoriya). It was interesting to see how the algorithm identified four groups without using predefined category labels (Viktoriya). We realized that clustering could help businesses recognize different customer groups based on spending habits (Viktoriya). This task helped us see how useful graphs can be when we want to understand what a machine learning algorithm is doing (Viktoriya).

In Task 3, we applied PCA to reduce the Iris dataset from four features to two principal components (Zaid). We created a scatter plot to see the results and found that the first two components kept approximately 97.77% of the total variance (Zaid). This was surprising because PCA reduced the number of dimensions by half while keeping almost all the original variation

(Zaid). Before this task, we did not fully understand how PCA worked, but now we know that it creates new components by combining information from the original features instead of simply removing columns (Zaid). This showed us how PCA can make a dataset easier to work with while preserving most of the original variation (Zaid). However, we were curious about how much information we would lose if we reduced the dataset to only one component instead (Zaid).

In the Knowledge Check, we answered questions that helped us review the main ideas behind unsupervised learning (Tien, Viktoriya, and Zaid). We explained the difference between supervised and unsupervised learning, especially how one uses target labels while the other looks for patterns without them (Tien). We discussed why choosing the lowest inertia does not always give us the best number of clusters (Viktoriya). We also explained what the high explained variance tells us about the Iris dataset (Zaid). These questions encouraged us to think about the meaning of our results instead of focusing only on completing the code (Tien, Viktoriya, and Zaid).

This lab connected to our previous work with Model Evaluation, the Bias-Variance Tradeoff, and data preprocessing (Viktoriya). In earlier labs, we compared predictions with actual values and examined how model complexity affected performance (Tien, Viktoriya, and Zaid). This time, we explored techniques that do not require target labels (Tien). The Elbow Method reminded us that increasing complexity does not always make a model more useful, while PCA introduced another way to work with features beyond cleaning and scaling them (Zaid). We now understand that choosing the right approach depends on what we want to learn from the data (Viktoriya). In future experiments, we would like to try these techniques on more complicated datasets to see how well they handle overlapping groups and larger numbers of features (Tien, Viktoriya, and Zaid).

In conclusion, this lab helped our group understand how unsupervised learning can discover patterns and simplify datasets without using target labels. K-Means Clustering showed us how to identify groups of similar points, while the Elbow Method helped us choose a reasonable number of clusters. PCA demonstrated how we could reduce dimensions while preserving most of the original variance. Most importantly, we learned to pay attention to what the graphs and results actually tell us rather than focusing only on running the code. This experience changed how we think about machine learning because we now understand that it can do more than predict outcomes.